

Title

Companion Care: Revolutionizing Early to Mid-Stage Dementia Care Through an Adaptive Artificial Intelligence Platform

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Background

Dementia affects millions of older adults in the United States and creates growing challenges for patients, caregivers, and healthcare systems. Individuals with early- to mid-stage dementia often experience changes in memory, behavior, mood, and daily functioning between clinic visits. Current dementia care often relies on episodic appointments, fragmented communication, and static educational resources, making it difficult to recognize patterns, respond early, and provide personalized support.

Objective

This project will develop Companion Care, an artificial intelligence (AI)-enabled companion care application designed to support individuals with early- to mid-stage dementia, their caregivers, and healthcare providers through personalized guidance, symptom tracking, and coordinated communication.

Approach/Methods

Companion Care will integrate patient- and caregiver-generated data through brief daily voice or tap-based caregiver logs, symptom tracking, personalized reminders, and caregiver support tools. Adaptive AI will identify individual patterns, or “trigger fingerprints,” related to behavioral and cognitive changes over time. Based on these patterns, the app will provide tailored support, including timely calming cues, dementia-related guidance, multilingual resources, and caregiver education. The platform will also generate concise physician-facing summaries that organize key changes, dates, and trends between visits without replacing clinical judgment. The design will be informed by caregiver and healthcare team interviews, literature review, and evaluation of existing dementia support applications.

Expected Outcomes

Companion Care is expected to improve symptom monitoring, caregiver support, care coordination, and clinical decision-making. By integrating daily home-based observations with adaptive AI, the platform will help identify emerging concerns earlier and support more personalized, proactive dementia care.

Conclusion/Impact

Companion Care will offer an innovative, multilingual, data-integrated approach to dementia care by connecting patients, caregivers, and clinicians through one shared platform. This tool will improve current approaches by moving beyond static resources toward continuous, coordinated support that enhances quality of life and continuity of care.